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Research Program: Brown Rice Chips with Herbs: (Basil, Onion Leaves, Parsley)	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☒ Food Research/ Food Development ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
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(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

Brown rice is some traditional health food rich in various active substances, which have effective preventive and therapeutic effects on many diseases. Besides basic nutrients, brown rice also contains a variety of bioactive ingredients, such as phytosterols, GABA, oryzanol, tocotrienols, vitamins, minerals, dietary fiber, functional lipids, phenolic acids, and essential amino-acids and phytic acid. It has anti-oxidation property, lipid-lowering functions, and shows potential in anti-diabetes, anti-inflammation, and enhanced immune activity (Ravichanthiran et al., 2018).

The study formulated the brown rice chips with herbs (basil, onion leaves, parsley), specifically to evaluate its sensory qualities and acceptability in terms of appearance, aroma, color, taste and crispiness. Finally, the products were submitted for microbial and proximate analysis of the best three treatments.

The method used in this study was developmental-experimental method of research. In the developmental research, this method used for formulation of brown rice chips with herbs for potential development and commercialization while in the experimental method attempted to investigate the proportion of brown rice chips with herbs using three treatments. This used the Completely Randomized Design (CRD), three products with different treatments were subjected to two trials: one (1) was tested by panel of evaluators and second (2) for final processes for consumer's preference evaluation by the 100 tasters. Score cards with the Nine (9) Points Hedonic Scale was used to obtain the data. The mean and Analysis of Variance (ANOVA) were used to analyze the data into alpha level set at 0.01 alpha. Findings on the sensory evaluation of the brown rice chips with herbs showed that Treatment B (onion leaves) was the best in all three quality attributes. Upon testing the microbial and proximate analysis of the best product, it was found out that there was a substantial amount of nutrients needed for human body. The brown rice chips with herbs was safe for human consumption as the results of microbial analysis of the product and based on the BFAD standard for microorganism test for products belonging to the Snack Foods category.

Keywords: brown rice, herbs, basil, onion leaves, parsley

(10) Introduction

In recent years, people have become increasingly interested in adopting a healthy diet, which also extends to healthy snacks, such as chips. The demand for high quality food has constantly increased during recent decades, as has the interest in the food quality issue both in response to market pressure such as requests from increasingly demanding and knowledgeable consumers and as a reaction to other factors, for example health and environmental concerns

Brown rice is some traditional health food It has anti-oxidation property, lipid-lowering functions, and shows potential in anti-diabetes, anti-inflammation, and enhanced immune activity. On the other hand, herbs such basil, onion leaves and parsley do not only enhance flavor but also contribute potential health benefits due to their antioxidant and micronutrient profiles. Parsley is low in calories yet rich in important nutrients, such as vitamins A, K, and C. Conversely, the only substantial nutrient that basil provides is vitamin K, but it also supplies plant compounds, which contribute aroma, flavor, and health benefits.

Therefore, this study endeavors to address this gap by investigating the acceptability of brown rice chips enhanced with herbs among consumers. By conducting a systematic evaluation

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of sensory attributes and consumer preferences, this research aims to provide valuable insights into the potential market viability and consumer appeal of herb-infused brown rice chips.

This study generally aimed to:

1. describe the sensory qualities of brown rice chips with herbs (basil, onion leaves, parsley) in terms of appearance, aroma, color, taste and crispiness;
2. determine the general acceptability of the product among three treatment groups;
3. determine the shelf life of raw and fried products and,
4. submit the best product for microbial and proximate analysis,.

Brown rice contains a high amount of dietary fiber, which has been shown to protect against colorectal cancer and breast cancer. In an animal study, rice bran from brown rice was shown to be beneficial against the development of polyps in the bowel. Due to its high content in fiber, brown rice has a lower glycaemic index, compared to white rice. Consumption of brown rice compared to white rice results in improved endothelial function, without changes in HbA1c levels, possibly through reducing glucose excursions. Vitamin E is also found in brown rice, mainly in two types of structure, which are tocopherols (α , β , γ and δ forms) and tocotrienols (α , β , γ and δ forms). The function of vitamin E is antioxidant activity, maintenance of membrane integrity, DNA repair, immune support, and metabolic processes (Aune et al., 2013).

There are two types of brown rice, which are germinated and non-germinated. Germinated brown rice is obtained by immersing the brown rice grain in water to initiate germination. The benefits of germinated brown rice are that the nutrients found in brown rice are more easily digested and the texture of brown rice is better. Germination has been employed to improve the texture of cooked brown rice. It also initiates numerous changes in the composition and chemical structure of the bioactive components. Germination could induce the formation of new bioactive compounds, such as gamma-aminobutyric acid (GABA). The consumption of germinated brown rice is increasing in many Asian countries because of its improved palatability quality and potential health-promoting functions (Wu et al., 2013).

Basil is used in traditional Tamil and Ayurvedic medicine, which is a form of traditional medicine popular on the Indian subcontinent. There are a number of types of basil, which differ in taste and smell. Sweet basil (the most commercially available basil used in Italian food) has a strong clove scent because of its high concentration of the chemical agent eugenol. Alternatively, lime and lemon basil have a strong citrus scent due to their high concentration of limonene. Basil comes in many varieties. While it may not contribute significant vitamins and nutrients to your diet, it can add unique flavor to your sauces, salads, and soups. Basil is a flavorful, leafy green herb that originated in Asia and Africa. It's a member of the mint family, and many different varieties exist. Popular as a food seasoning, this aromatic herb is also used in teas and supplements which may provide a range of health benefits. A member of the mint family (Lamiaceae), basil is one of the most popular culinary herbs. It's a warm-weather annual herb you can plant outdoors once temperatures are consistently above 50°F. Basil is not usually seeded directly into the soil; typically, gardeners transplant small starter plants purchased at a nursery. Or, they may start the seeds themselves indoors under grow lights.

Onion Leaves is the most widely cultivated vegetable around the world. In this study, the isolation, concentration, quantification, and bioactivity evaluation of the phenolics in onion peels were investigated. Thirty-four phenolics, including 17 flavonoids and 17 nonflavonoid phenolics, were purified and identified. Among them, there were 2 new unusual epoxyflavanones and a new

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phenolic constituent, as well as 13 unreported constituents from the genus *Allium*. Rosemary is a member of the mint family *Lamiaceae*, along with many other herbs, such as oregano, thyme, basil, and lavender. The herb not only tastes good in culinary dishes, such as rosemary chicken and lamb, but it is also a good source of iron, calcium, and vitamin B-6. It is typically prepared as a whole dried herb or a dried powdered extract, while teas and liquid extracts are made from fresh or dried leaves. The herb has been hailed since ancient times for its medicinal properties. Rosemary was traditionally used to help alleviate muscle pain, improve memory, boost the immune and circulatory system, and promote hair growth.

Parsley is a debasement of two Latin words, 'petros' and 'selinon' signifying 'shake' and 'celery', individually. Dioscorides, a Greek doctor of the early Roman Empire, is said to have given the plant its name, 'Petroselinum crispum'. Parsley initially developed wild close to the Mediterranean. Parsley was acknowledged for its restorative properties well before it wound up acknowledged as a nourishment or flavor. Since parsley was mistaken for false parsley (which is noxious), the level assortment rushed to get on. Parsley can clean your breath and sense of taste. Parsley is a brilliant green bare biennial herbaceous plant in mild atmospheres, and a yearly herb in subtropical regions. Parsley is wealthy in iron, nutrients A, B and C and oil called apiol, which is separated from the seeds and used to treat contaminations of the urinary tract, general infection anticipation, assimilation and course issues, and kidney stones (Ajmera et al., 2019).

Related Studies:

Kroepeck from Cagaycay (*Venerupis philippinarium*) is a food product locally known as "kroepeck" and is mainly produced from meat of cagaycay (*Venerupis philippinarium*), an endemic shellfish species in the Visayas area of the Philippines. The kroepeck of the present utility model has the following composition: ground rice: 23.01 pcnt -33.52 pcnt, flour: 23.01 pcnt -33.52 pcnt, cagaycay meat: 11.28 pcnt -22.05 pcnt, alogbate (*Basel/a spp.*) leaves: 4.41 pcnt -5.59 pcnt, saluyot (*Corchorus spp.*): 4.41 pcnt -5.59 pcnt, okra (*Abelmoscusesculentus*): 4.41 pcnt - 5.59 pcnt, ginger: 1.37 pcnt -3.02 pcnt, bell pepper: 2.20 pcnt -2.79 pcnt, and spices: 0.50 pcnt -7.85 pcnt (Nava et al. 2018).

A Process of making Kroepeck using Pigeonpea Flour (*Cajanus Cajan L*) A process of making Kroepeck with pigeonpea flour comprising of the steps of: preparing ingredients namely 1-2 cups rice flour, 0.25-1 cup pigeonpea flour, 0.5-1 teaspoon garlic powder, 0.5-1 cup water, and flavoring; mixing all the measured ingredients until homogenous mixture is attained; kneading the dough until smooth and elastic; flattening the dough in the pan or tray until attaining the desired thickness; steaming the dough within 4 to 5 mins; removing steam dough and cutting into strips of desired shape using knife or cutter; sun drying the dough until the dough becomes brittle; deep frying the dried dough in a cooking oil until attaining desired texture and crispiness; letting it cool; and packing in polyethylene bag and seal using bag sealer (Bernabe et al., 2020).

A Method for Preparing Crackers From Eragrostis Tef Flour The present disclosure relates to a method for preparing crackers from Eragrostis tef flour. The method comprises mixing Eragrostis tef flour with a liquid medium and allowing the obtained batter to stand to obtain a fermented batter. The fermented batter is then uniformly spread to a wall thickness in the range of 3 mm to 5 mm on a heated griddle to obtain a crepe with random air pockets. The crepe is placed between a pair of heated discs and an incremental axially directed pressure is applied to reduce the wall thickness in the range of 0.5 mm to 1.5 mm to obtain the cracker. The crackers prepared by the method of the present disclosure retain the nutritional properties,

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are palatable, easily chewable and can be used as a healthy addition to the routinely used diet (Mehta et al., 2019).

(11)

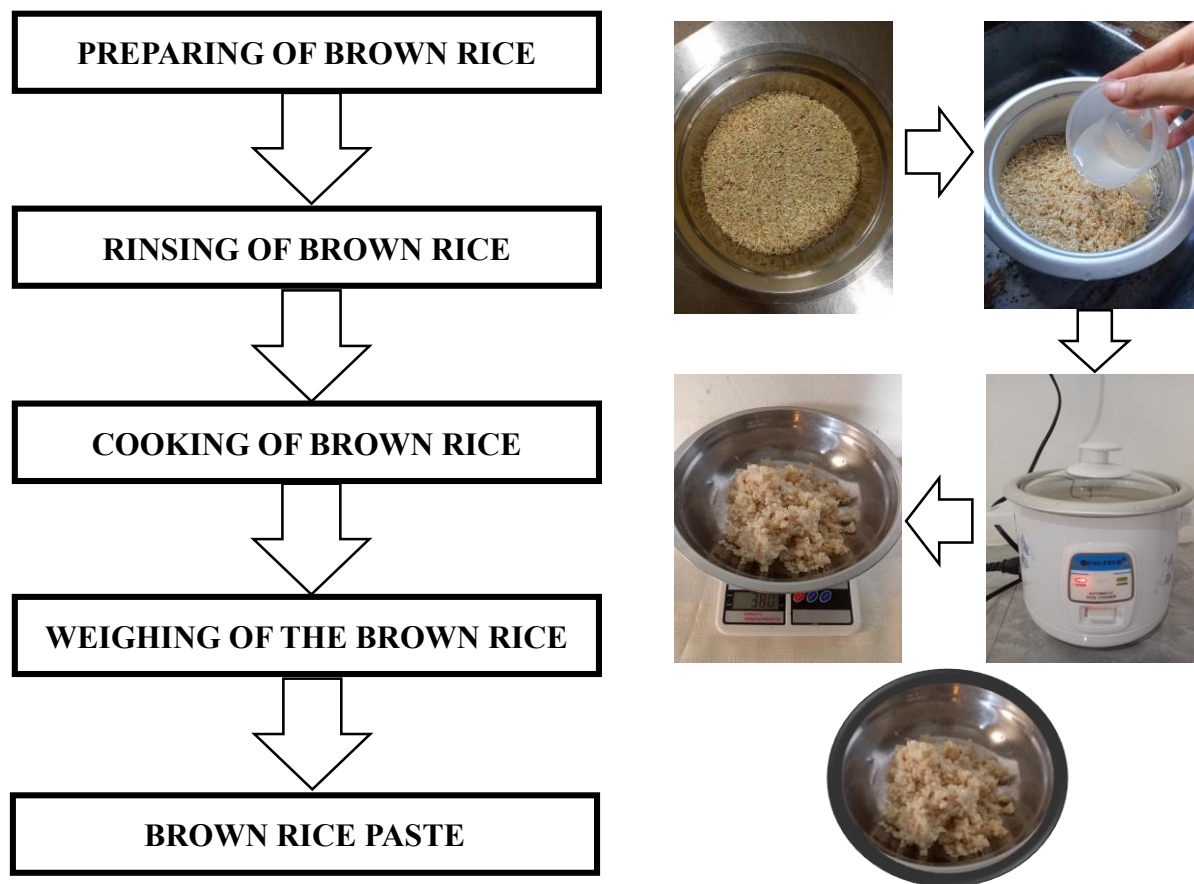
The method used in this study was developmental-experimental method of research. In the developmental research, this method used for formulation of brown rice chips with herbs for potential development and commercialization while in the experimental method attempted to investigate the proportion of brown rice chips with herbs using three treatments. This used the Completely Randomized Design (CRD), three products with different treatments were subjected to two trials: one (1) was tested by panel of evaluators and second (2) for final processes for consumer's preference evaluation by the 100 tasters. Score cards with the Nine (9) Points Hedonic Scale was used to obtain the data. The mean and Analysis of Variance (ANOVA) were used to analyze the data into alpha level set at 0.01 alpha.

Step 1. Preparing of Raw Materials. The raw materials needed for the developmental of Brown Rice Chips with Herbs (basil, onion leaves, parsley) are gathered and inspected to ensure good quality.

a. Preparation for Brown rice. A brown rice was purchased. Then rinsed in clean running water and cooked. Using the weighing scale weighed/measured the brown rice. Set it aside.

b. Preparation of basil, onion leaves, parsley.

Basil, Onion leaves, Parsley was purchased at Bagong Lipunan Trade Center from a known vendor. The leaves are washed in clean running water. Then it was slice thinly. Set aside.

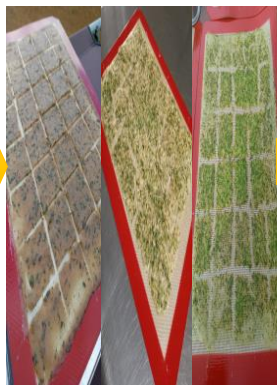


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Step 2. Procedures in Making Brown Rice Chips with Herbs (basil, onion leaves, parsley).

The experimental procedure for producing Brown Rice Chips with Herbs (basil, onion leaves, parsley) involves several steps,

First, gather all necessary ingredients, including brown rice paste, basil, onion leaves, parsley, water, salt, and oil. In a large mixing bowl, combine the chopped herbs with the brown rice paste then add water and salt to the mixture and mix well to ensure even distribution of moisture. Blend for about 4-5 minutes until it forms into a paste. Then arrange on a silicon mat using spatula, and slice using dough cutter and dry under the sun for 5 hours until crisp. Preheat the oil to the desired temperature for frying the chips. Remove the chips from the pan and allow them to cool this will help them firm up and become crispier and stored in a cool and dry place. Note that the crispness of the chips depends on how the chips was being stored or it may have lost the crispiness.



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1. Production of technology manual/cooking guide/GMP manual

The quality standards, cooking guide, and other IEC materials will be developed and produced. It will be disseminated and distributed to various interested stakeholders.

The experimental design will be CRD with 3 treatments. The gathered data will be computed using the SPSS and analyzed using Analysis of Variance (ANOVA) at a 0.01 percent level of significance.

(12) Results and Discussions

Sensory Evaluation of Brown Rice Chips with Herbs

The sensory evaluation of brown rice chips infused with varying amounts of herbs (basil, onion leaves, and parsley) was conducted by a panel of experts, assessing key attributes such as appearance, aroma, color, taste, and crispiness, as presented in Table 1. In general, the sensory evaluation results suggest that while all variations of brown rice chips infused with basil, onion leaves, and parsley offer appealing sensory attributes, preferences may vary based on individual taste preferences and the specific herb content.

Regarding appearance, brown rice chips containing 30 g of herbs received high ratings across all three herb varieties, with mean scores ranging from 7.30 to 7.90, indicating a generally strong appeal in terms of visual presentation. Similarly, the aroma of these chips was deemed very pleasant, with mean scores ranging from 7.30 to 7.60, reflecting the enticing fragrance imparted by the herbs.

In terms of color, chips containing 30 g of onion leaves and parsley were rated as very authentic, while those with basil were deemed moderately authentic. This suggests that onion leaves and parsley contribute more significantly to the chips' color authenticity compared to basil.

Moving on to taste, chips with 30 g of basil received a rating of moderately delicious, while those with onion leaves and parsley were deemed very delicious. This indicates a preference for the taste profile imparted by onion leaves and parsley over basil.

Furthermore, all variations of the brown rice chips, irrespective of the herb content, were found to be very crispy, indicating a desirable textural characteristic that adds to the overall sensory experience.

Table 1. Sensory qualities of brown rice chips with herbs

TREATMENTS		BASIL		ONION LEAVES		PARSLEY	
Product	Quality Attributes	Mean	AD	Mean	AD	Mean	AD
30 g	Appearance	7.80	VMA	7.30	VMA	7.90	VMA
	Aroma	7.30	VMP	7.60	VMP	7.50	VMP
	Color	6.90	MA	7.40	VMA	7.80	VMA
	Taste	7.00	MD	7.50	VMD	7.70	VMD
	Crispiness	7.50	VMC	7.80	VMC	7.50	VMC

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40 g	Appearance	7.50	VMA	7.70	VMA	7.90	VMA
	Aroma	7.90	VMP	7.70	VMP	7.60	VMP
	Color	7.80	VMA	7.60	VMA	7.50	VMA
	Taste	7.40	VMD	7.70	VMD	7.20	MD
	Crispiness	7.70	VMC	7.80	VMC	7.60	VMC
50 g	Appearance	7.60	VMA	7.90	VMA	7.20	MA
	Aroma	7.60	VMP	7.50	VMP	7.40	VMP
	Color	7.90	VMA	7.70	VMA	7.20	MA
	Taste	7.30	VMD	8.10	VMD	6.70	MD
	Crispiness	8.00	VMC	8.00	VMC	8.00	VMC

General Acceptability of Brown Rice Chips with Herbs as evaluated by consumers

The findings presented in Table 2 shed light on the overall acceptability of brown rice chips infused with herbs (basil, onion leaves, parsley) as perceived by a group of consumers, focusing on sensory qualities associated with the 30 g component composition of these herbs within the chips. Notably, Treatment B, featuring onion leaves, emerged as the most favored option across all evaluated sensory attributes, garnering impressive mean scores of 8.51 for appearance, 8.35 for aroma, 8.38 for color, 8.51 for taste, and 8.62 for crispiness. These high ratings collectively led to treatment B being described as "Liked Extremely" for all five sensory qualities, indicating a robust level of general acceptability among consumers.

Table 2. General Acceptability of brown rice chips with herbs as evaluated by consumers

Treatments	A (Basil)		B (Onion Leaves)		C (Parsley)	
<i>Quality Attributes</i>	<i>Mean</i>	<i>AD</i>	<i>Mean</i>	<i>AD</i>	<i>Mean</i>	<i>AD</i>
Appearance	7.60	LVM	8.51	LE	8.12	LE
Aroma	7.59	LVM	8.35	LE	7.93	LVM
Color	7.69	LVM	8.38	LE	8.09	LVM
Taste	7.24	LVM	8.51	LE	8.02	LVM
Crispiness	7.87	LVM	8.62	LE	8.11	LVM
Acceptability	7.60	LVM	8.47	LE	8.05	LVM

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Legend: **Adjectival Description (AD)**

Score	General Acceptability
8.12 – 9.00	<i>Liked Extremely (LE)</i>
7.23 – 8.11	<i>Liked Very Much (LVM)</i>
6.34 – 7.22	<i>Liked Moderately (LM)</i>
5.45 – 6.33	<i>Liked Slightly (LS)</i>

Shelf-life of the Raw Brown Rice Chips with Herbs

The shelf-life of raw brown rice chips with various herb treatments at room temperature. The variants include Treatment A with basil, Treatment B with onion leaves, and Treatment C with parsley. The evaluation criteria are mold formation and the presence of an unpleasant odor. The table further shows that all treatments (A, B, and C) showed no mold formation and no unpleasant odor at the one-month mark. At the two-months mark, Treatment A (basil) and Treatment B (onion leaves) Treatment C (parsley) showed mold formation, indicated by the presence of spots, and an unpleasant odor at the two-month mark.

Table 5. Shelf-life of the raw brown rice chips with herbs

Variant	One-month Mold Formation, unpleasant odor	Two-months Mold Formation, unpleasant odor, production of spot
Treatment A	0	+
Treatment B	0	+
Treatment C	0	+

Legend: 0 – absence of molds + – presence of molds

Shelf-life of the Fried Brown Rice Chips with Herbs Room Temperature

The shelf-life of fried brown rice chips with various herb treatments at room temperature. Similar to the previous table, the variants include Treatment A with basil, Treatment B with onion leaves, and Treatment C with parsley. The evaluation criteria remain mold formation and the presence of an unpleasant odor. The table further showed that all treatments (A, B, and C) showed no mold formation and no unpleasant odor from one to six days. From seven to fourteen days, all treatments continued to exhibit no mold formation and no unpleasant odor. At fifteen days, Treatment A (Basil) Treatment B (Onion Leaves) Treatment C (Parsley) exhibited mold formation, indicated by the presence of spots, and an unpleasant odor at the fifteen-day mark.

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Table 6. Shelf-life of the fried brown rice chips with herbs

Variant	One to Six Days Mold Formation (unpleasant odor)	Seven to Fourteen Days Mold Formation (unpleasant odor)	Fifteen Days Mold Formation (unpleasant odor, production of spot)
Treatment A (Basil)	0	0	+
Treatment B (Onion Leaves)	0	0	+
Treatment C (Parsley)	0	0	+

Legend: 0 – absence of molds

+ – presence of molds

Microbial Analysis of Brown Rice Chips with Onion Leaves

The Bureau of Food and Drugs standard for microorganism test of the products. Therefore, to lower the molds and yeast formation, it is recommended that the product must undergo safe and sanitized preparation, well-packaged and stored in the room temperature.

The findings began in the sample tested which was brown rice chips with onion leaves, with a quantity of 200 grams. The parameters tested include the Aerobic Plate Count which indicates the total number of aerobic bacteria present in the sample. The Total Coliform Count Measures the presence of coliform bacteria, which can indicate potential contamination. Then the *Escherichia coli* (E. coli) Count, which specifically targets the presence of E. coli bacteria, which can be harmful if present in food. Lastly, the Molds and Yeast Count which assesses the presence of mold and yeast, which can affect the quality and safety of the product.

Table 7. Microbial analysis brown rice chips with onion leaves.

Sample Description	Parameter	DOST Result	FDA M	M
Brown Rice Chips with Onion Leaves 200g	Aerobic Plate Count	< 250 cfu/g sample	10 ³	10 ⁴
		< 10 cfu/g sample		
	Total Coliform Count		10	10 ²
		< 10 cfu/g sample		
	<i>Escherichia coli</i> Count	< 100 cfu/g sample	10	10 ²
	Molds and Yeast Count			

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Brown rice can be utilized as the main ingredients in making chips. Moreover, all three treatments displayed distinctive sensory characteristics in which all variants exhibited a visually appealing appearance, with the presence of herbs contributing to the visual appeal. Each treatment imparted a unique aroma to the chips, with basil offering a fresh and herbaceous scent, onion leaves providing a pungent aroma, and parsley adding a fresh, grassy aroma and the chips maintained their natural brown color, with slight variations depending on the herb treatment. Likewise, basil-treated chips had a subtle herbaceous flavor, onion leaf-treated chips offered a more pronounced onion flavor, and parsley-treated chips provided a fresh, slightly bitter taste. Overall, all variants exhibited a desirable level of crispiness, enhancing the overall texture of the chips and the basil, onion leaves and parsley can be used as added ingredients in making a chip.

The general acceptability of the product was assessed among the three treatments. Overall, all treatments were well-received by the panelists, indicating a positive perception of the brown rice chips with herbs. However, individual preferences may vary based on personal taste preferences. Moreover, based on the results as evaluated by panel of expert. Among the three (3) treatments of brown rice chips with herbs, Treatment C has the best quality attributes. It has the best proportion among all treatments and was very liked by the evaluators in terms of its appearance, aroma, color, taste and crispiness.

Recommendations

Based on the conclusions of the study, the following recommendations are drawn:

It is recommended that more experiment with different formulations and concentrations of herbs to optimize flavor balance while maintaining product integrity and consider incorporating additional herbs or spices to offer a wider variety of flavor options and cater to diverse consumer preferences then explore alternative cooking methods or ingredient combinations to enhance the overall sensory experience of the chips.

Highlight the natural and healthy attributes of the product, emphasizing the use of whole grains and natural herbs in the chip formulations. Leveraging unique flavor profiles of each herb treatment to differentiate the product in the market and attract consumers seeking innovative snack options and collaborate with nutritionists or health experts to communicate the nutritional benefits of brown rice chips and herbs to health-conscious consumers.

Provide detailed information on the health benefits and culinary uses of the herbs used in the chip formulations to educate consumers and enhance product appeal and offer recipe ideas or serving suggestions that showcase the versatility of brown rice chips with herbs, encouraging consumers to incorporate them into their daily meals or snacks.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible).

Brown rice chips with herbs can be produced and can be commercialized

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

The product of this study could be one of the sources of extension activities conducted by the University, particularly on the main campus.

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(16) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(19) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal,

Researchgate <https://www.researchgate.net/>

Mendeley <https://www.mendeley.com/>

Linkedin <https://www.linkedin.com/>

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